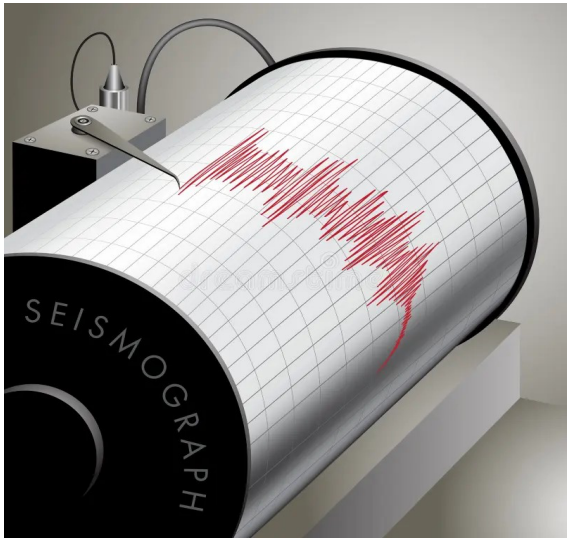
The seismometer

The seismometer is a scientific instrument used for detecting and recording the movements of the ground caused by earthquakes. It is essential in the study of seismic phenomena, because it allows researchers to analyse the intensity, duration and direction of seismic waves.

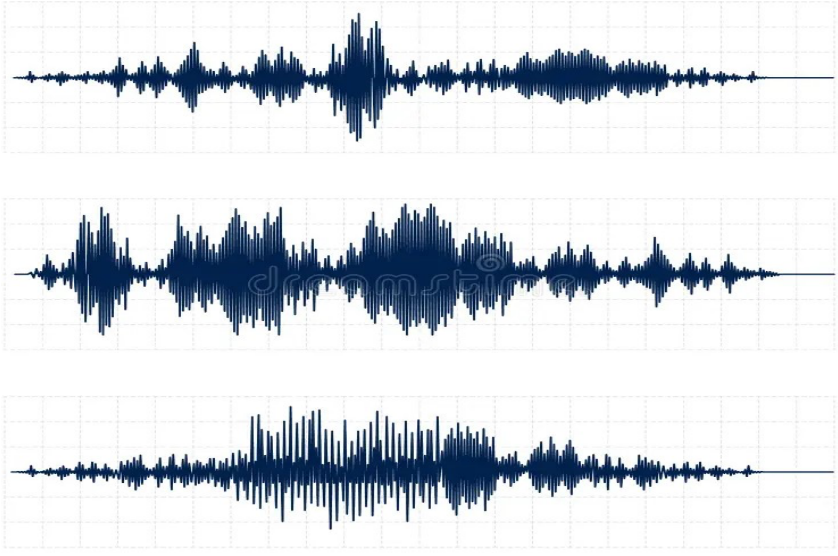


Seismometer

The way a seismometer works is based on the principle of inertia. The device is made up, in simplified terms, of a suspended mass (which remains relatively still during an earthquake) and a support fixed to the ground, which moves together with the ground.

At the moment when an earthquake occurs, the ground moves but the mass tends to stay in place, and this difference in movement is recorded in the form of signals.

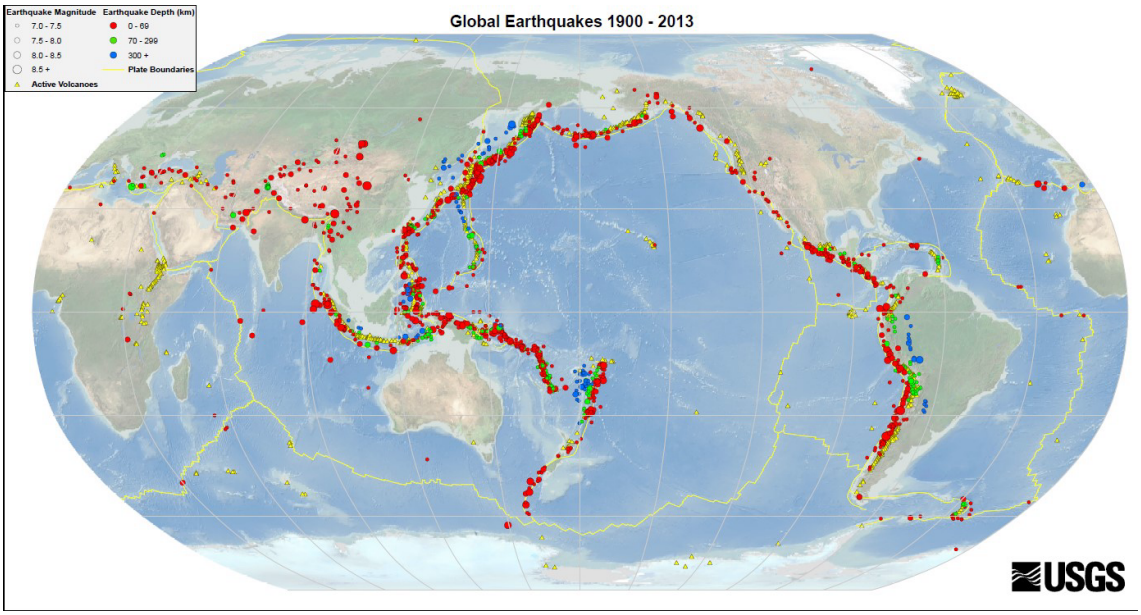
In the past, these movements were drawn on paper as lines called seismograms. Nowadays, modern seismometers use very precise digital systems that can transmit data in real time to seismic monitoring centres.



Example of a seismogram — recorded ground motion.

Seismometers are placed in different areas around the globe, especially in regions with high seismic activity. Networks of seismometers help with quickly locating the epicentre of an earthquake and with assessing its magnitude.

The importance of the seismometer is very great, because it contributes to our understanding of how earthquakes occur and to the development of early-warning systems. Therefore, although they cannot prevent earthquakes, these instruments help to reduce the risks and to protect human lives.



Distribution of major global earthquakes (1900–2013) and tectonic-plate boundaries (USGS).